

Nathan Lemma

Software Engineer · Systems, AI & GPUs · Austin, TX

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Education

The University of Texas at Austin — Austin, TX Aug 2023 — May 2027
Bachelor of Science: *Electrical and Computer Engineering*
Coursework: *Distributed Systems (Grad) | Systems for ML (Grad) | Multithreading & GPUs | Operating Systems | Computer Architecture*

Publications

SOSP '26 · EnerTune · “Beyond Utilization: Energy-Conscious GPU Sharing for Inference Serving” 3rd author
doi:10.1145/3830418.3843913 | github.com/UT-SysML/EnerTune

Experience

UT Austin Research · Undergraduate Research Assistant — Austin, TX Mar 2024 — Present
Machine Learning for Systems: LDOS Lab · Akella Aug 2026 — Present

- Building a harness streaming live vLLM scheduler and memory stats to an LLM agent tuning vLLM’s cross-replica request router

Systems for Machine Learning: SysML Lab · Yadwadkar Aug 2025 — May 2026

- Rebuilt 4 state-of-the-art GPU-sharing inference serving systems (GPUlets, Usher, FGD, ParvaGPU) to benchmark EnerTune
- Debugged and ported each baseline’s research prototype to run on our 16-A100 cluster, enabling the 1.4–2.3× lower energy result
- Profiled 4 baselines to measure EnerTune’s 7.3× average (17.3× max) profiling speedup over brute-force and kernel-level methods

Computer Architecture and FPGAs: Chiou Group Mar 2024 — Dec 2024

- Built an instruction-accurate RISC-V simulator in C++ so the team could test the “Primate” compiler without slow FPGA synthesis
- Designed I/O instruction handling and compiler-checking tools that sped up developing the hardware and software side by side

Caterpillar Inc. · Software Engineering — Peoria, IL May 2024 — Present
Software Intern: Data Acquisition and Analytics Team Summer 2026, Part-Time through 2026–27

- Building cellular-cost tracking for 20,000+ machines from 100 GB/day, cutting attribution error by 34% on average (up to 55%)
- Deploying a nightly AWS Lambda that invokes carrier APIs to locate each machine, aggregate usage, and calculate its actual cost
- Assembling an ASP.NET dashboard with cross-source SQL joins and caching, giving every engineering team daily cost visibility

Systems Intern: System Diagnostics Team Summer 2025, Part-Time through 2025–26

- Wrote hardware-in-the-loop tests covering 30 I/O types (PWM, analog, and digital) to set a baseline for the whole diagnostic suite
- Translated hardware requirements into embedded test modules so every new feature automatically gets regression-test coverage
- Automated key steps in the firmware build process used daily by a team of 15+ engineers, cutting their configuration time by 10%

Embedded Intern: Platform Software Team Summer 2024, Part-Time through 2024–25

- Designed hardware-abstraction (HAL) and API layers for engine control modules (NVM, analog, and digital I/O), now in production
- Developed firmware components in C and Python that cut new I/O support time by 15%, verified via .elf memory-layout checks

Longhorn Racing (Formula SAE) · Battery Management Developer — Austin, TX Aug 2023 — May 2025
Embedded Systems

- Led the design of the car’s battery monitoring system (BMS); wrote RTOS firmware modules handling drivers, I/O, and scheduling
- Built the State-of-Charge algorithm that improved battery range estimates, cutting the driving-range margin of error to under 5%

Projects

Energy-Efficient MLLM Serving: Group Project, Systems for ML Grad Class (CUDA, vLLM) mlm-energy.github.io/MLLM-Energy

- Profiled multimodal LLM serving per phase on an A100, then 1,080 SM-count configurations for throughput, power, and energy
- Found decode accounts for 89% of runtime at batch size 1 despite averaging only 14% SM throughput and 23% DRAM throughput
- Found 67% fewer active GPU SMs reduced serving throughput by under 9% on output-heavy workloads across batch sizes 1–16
- Measured 13× lower energy per request when batching output-heavy jobs, but only 2× on input-heavy ones due to encoder cost

Concurrent Hash Tables & Radix Trees: Multi-threaded Data Structure Benchmarking (C/C++) code-tomato.github.io/CHT_Radix

- Built 7 concurrent data structures in C++ for the shared prefix-cache lookups that bottleneck multi-threaded LLM serving engines
- Achieved a 13× throughput gain on the same radix-tree structure by improving thread coordination, peaking at 88.36 million ops/sec
- Found roughly 40× p99 latency gaps between hash tables on a ShareGPT trace, showing structure choice strongly affects tail latency

Additional

Skills: C/C++, CUDA, Python, SQL, Bash | vLLM, gdb, QEMU, Docker, Git | AWS (Lambda), Oracle, MySQL | .NET, React, HTML/CSS, Tailwind
Awards: Samsung Scholarship, Santander Scholarship, Albert and Jessie Cudlipp Scholarship
Organizations & Leadership: Students4Hope (Founder), SEO USA, Ron Brown Scholar, ColorStack, NSBE, ABCS, OCCM